

LITCHFIELD, MN, USA

WMO: 726583

Lat: 45.147N

Lon: 94.507W

Elev: 348

StdP: 97.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.7	-29.1	0.3	-24.8	-27.0	0.3	-22.7	11.4	-11.2	10.3	-9.7	2.9	310	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.1	23.3	30.1	22.4	28.1	21.1	25.3	29.7	24.1	27.9	23.0	26.7	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.6	27.9	22.7	18.2	26.6	22.2	17.6	25.8	79.7	29.1	74.5	28.1	70.2	26.8	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.9	-28.7	34.7	3.3	1.6	-31.1	35.8	-33.0	36.7	-34.9	37.6	-37.3	38.8		
(5)			-29.1	27.2	3.2	1.7	-31.4	28.4	-33.2	29.4	-35.0	30.4	-37.3	31.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-10.2	-8.1	-0.8	7.1	14.3	19.9	22.4	20.9	16.9	8.9	0.9	-6.9
	DBStd	12.68	7.38	7.24	6.93	5.71	4.82	3.71	3.09	2.93	4.68	5.04	6.00	6.65
	HDD10.0	2490	625	507	339	119	15	0	0	0	4	81	277	523
	HDD18.3	4417	883	741	593	337	144	26	4	10	81	294	522	782
	CDD10.0	1468	0	0	4	33	147	297	385	337	212	48	5	0
	CDD18.3	351	0	0	0	1	18	72	131	89	38	2	0	0
	CDH23.3	2854	0	0	1	21	193	589	1102	621	303	24	1	0
	CDH26.7	821	0	0	0	4	59	169	352	154	78	3	0	0
Wind	WSAvg	3.3	3.4	3.5	3.6	4.0	3.7	3.1	2.5	2.3	2.8	3.4	3.4	3.3
Precipitation	PrecAvg	764	20	20	42	72	84	123	99	94	82	59	37	19
	PrecMax	975	51	51	89	179	171	203	188	192	208	123	101	42
	PrecMin	595	1	3	14	21	22	51	22	9	13	19	1	8
	PrecStd	116	15	13	21	39	37	39	42	40	48	30	32	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	11.3	19.1	26.3	32.2	33.0	33.9	32.4	31.4	26.4	18.9	8.9
		MCWB	3.2	8.8	13.6	15.5	19.8	22.6	24.9	24.0	22.0	16.8	13.0	7.2
	2%	DB	2.9	6.3	15.1	22.3	27.8	31.0	32.2	30.0	28.8	22.5	15.1	5.9
		MCWB	1.4	3.9	10.9	12.9	17.6	22.0	24.4	23.3	21.6	15.6	10.0	3.4
	5%	DB	2.0	3.7	12.1	18.9	25.3	28.3	30.2	27.9	27.2	19.1	12.2	2.8
		MCWB	1.1	2.3	9.2	11.5	16.2	20.4	23.2	21.7	20.0	13.6	8.4	1.7
	10%	DB	-0.1	2.2	7.9	16.4	22.5	27.1	28.1	27.1	24.7	17.2	8.9	2.0
		MCWB	-0.6	1.2	5.6	9.8	15.2	19.5	22.0	20.9	18.6	12.5	6.2	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	8.8	15.5	16.8	22.6	25.7	27.8	26.0	25.1	19.5	14.5	7.7
		MCDB	6.1	10.7	18.0	23.9	27.9	30.3	30.5	30.3	29.1	22.9	17.7	8.4
	2%	WB	2.0	4.1	11.9	14.3	20.0	23.7	25.8	24.6	23.0	16.9	11.2	3.4
		MCDB	3.2	5.6	14.6	20.0	24.6	28.0	30.1	28.3	26.4	20.2	14.4	4.9
	5%	WB	0.7	2.6	8.7	12.4	17.9	22.2	24.6	23.4	21.5	14.9	8.6	2.2
		MCDB	1.7	3.7	11.5	17.3	22.4	26.8	28.7	26.7	25.2	18.8	11.4	3.1
	10%	WB	-0.9	0.8	5.8	10.7	16.3	20.7	23.4	22.3	19.9	12.8	6.4	0.9
		MCDB	-0.3	2.1	8.2	15.7	20.9	25.0	27.2	25.3	23.7	16.3	9.1	1.8
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.2	9.1	10.4	10.6	10.1	10.1	9.9	10.8	9.5	8.3	7.6
		MCDBR	9.4	9.8	12.8	15.2	14.8	12.1	11.7	11.4	12.9	13.5	12.5	9.1
		MCWBR	7.9	7.9	8.6	8.1	6.8	5.8	6.1	5.9	6.5	7.5	8.1	7.4
	5% WB	MCDBR	9.0	9.0	12.3	13.1	12.0	10.5	10.3	10.0	11.1	11.1	11.5	8.2
		MCWBR	7.8	7.4	8.6	8.1	6.9	6.3	6.3	6.2	7.0	7.2	8.1	6.8
Clear-Sky Solar Irradiance	taub	0.260	0.277	0.307	0.363	0.391	0.390	0.389	0.401	0.363	0.314	0.282	0.257	
	taud	2.437	2.382	2.416	2.327	2.304	2.356	2.365	2.320	2.401	2.534	2.524	2.460	
	Ebn at Noon	872	916	929	897	878	879	874	849	856	864	839	839	
	Edh at Noon	69	89	100	119	126	121	118	119	101	77	64	61	
All-Sky Solar Radiation	RadAvg	1.76	2.83	3.77	4.67	5.39	6.00	6.44	5.35	4.14	2.65	1.70	1.25	
	RadStd	0.22	0.32	0.35	0.42	0.42	0.43	0.23	0.43	0.31	0.37	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	+0.75	N/A	N/A	N/A	N/A	N/A
Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page